

ECG Anomaly Detection — Session Report

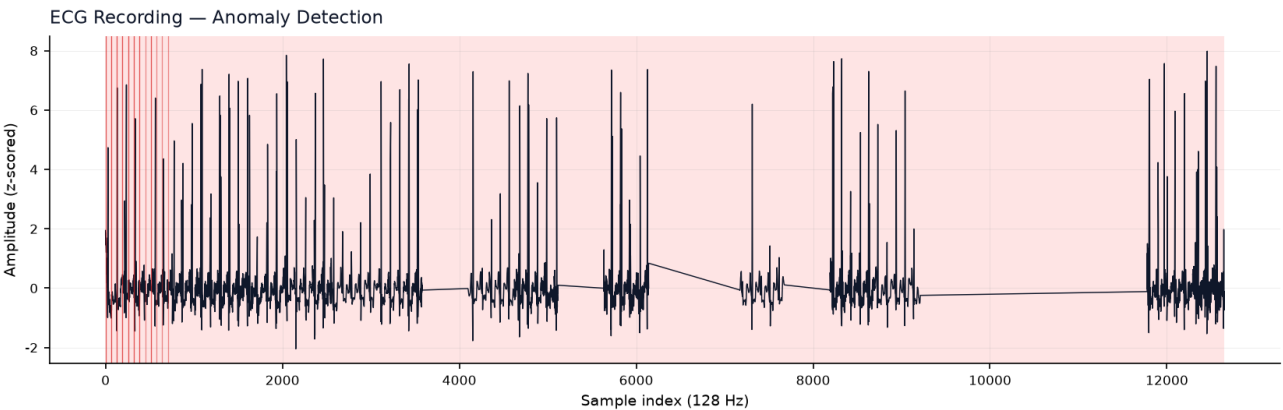
Prepared for: ss • Session ID: 0a7cc16b • Generated: 2026-07-04 18:11:35

Session Information

Patient / User	Sriyantha Sadaruwan
Source	mit-bih-arrhythmia (—)
Started	2026-07-04 18:11:35
Ended	2026-07-04 18:12:32
Duration	56.7 s
Status	completed
Total beats analyzed	28,672
Anomalous beats	19,999 (69.75%)
Model	ecg-ae-2026-07-04 (v1783187890) • Dilated-U-Net-Autoencoder
Detection threshold (MAE)	0.1006

ECG Recording & Detection Results

The following plot shows the recorded ECG signal (z-score normalized). Red-shaded regions indicate samples flagged as anomalous by the autoencoder (reconstruction error exceeding the trained threshold).



Alerts (342)

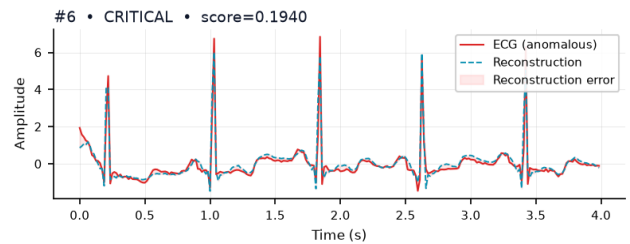
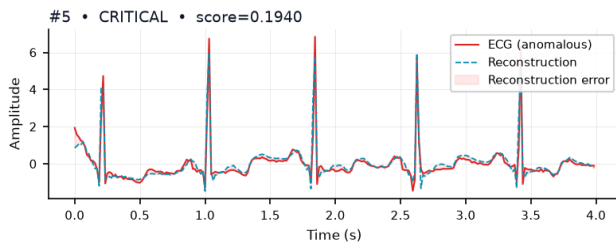
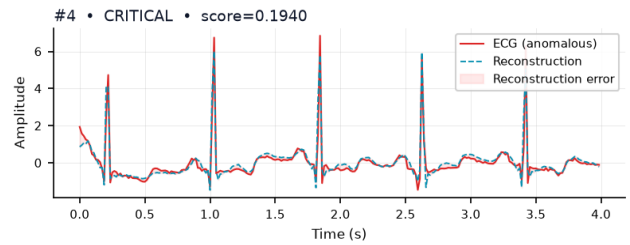
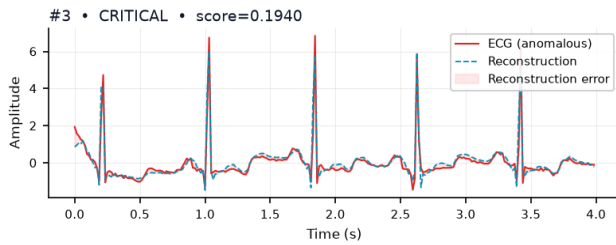
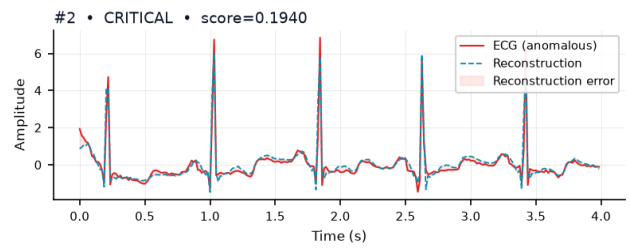
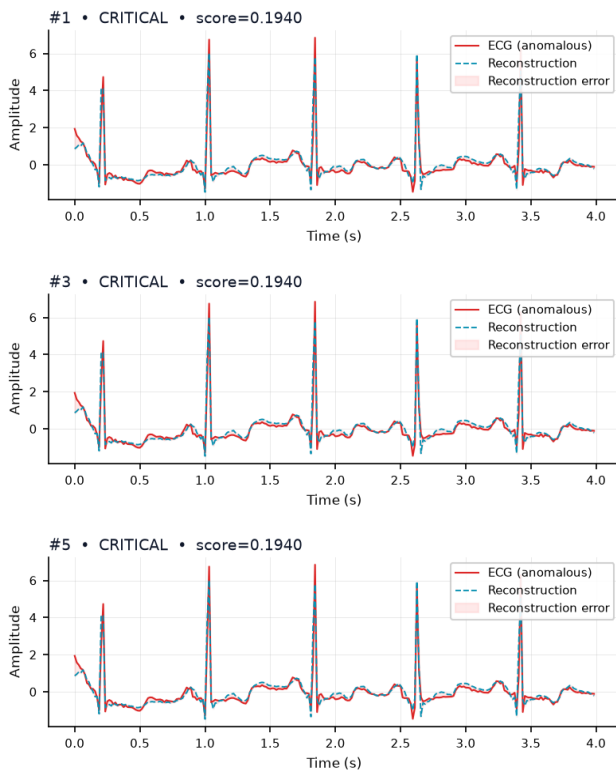
Time	Severity	Score	Threshold	Message
2026-07-04 18:11:36	critical	0.1940	0.1006	Premature Atrial Contraction detected (score=0.1940, confidence=0.9999)
2026-07-04 18:11:37	critical	0.1940	0.1006	Premature Atrial Contraction detected (score=0.1940, confidence=0.9999)
2026-07-04 18:11:37	critical	0.1940	0.1006	Premature Atrial Contraction detected (score=0.1940, confidence=0.9999)
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2026-07-04 18:11:37	critical	0.1940	0.1006	Premature Atrial Contraction detected (score=0.1940, confidence=0.1006)
2026-07-04 18:11:37	critical	0.1685	0.1006	Premature Atrial Contraction detected (score=0.1685, confidence=0.1006)
2026-07-04 18:11:37	critical	0.1940	0.1006	Premature Atrial Contraction detected (score=0.1940, confidence=0.1006)
2026-07-04 18:11:38	critical	0.1685	0.1006	Premature Atrial Contraction detected (score=0.1685, confidence=0.1006)
2026-07-04 18:11:38	critical	0.1685	0.1006	Premature Atrial Contraction detected (score=0.1685, confidence=0.1006)
2026-07-04 18:11:38	critical	0.1744	0.1006	Premature Atrial Contraction detected (score=0.1744, confidence=0.1006)
2026-07-04 18:11:39	critical	0.1685	0.1006	Premature Atrial Contraction detected (score=0.1685, confidence=0.1006)
2026-07-04 18:11:39	critical	0.1744	0.1006	Premature Atrial Contraction detected (score=0.1744, confidence=0.1006)
2026-07-04 18:11:39	critical	0.1685	0.1006	Premature Atrial Contraction detected (score=0.1685, confidence=0.1006)
2026-07-04 18:11:39	critical	0.1744	0.1006	Premature Atrial Contraction detected (score=0.1744, confidence=0.1006)
2026-07-04 18:11:39	critical	0.1744	0.1006	Premature Atrial Contraction detected (score=0.1744, confidence=0.1006)
2026-07-04 18:11:39	critical	0.1685	0.1006	Premature Atrial Contraction detected (score=0.1685, confidence=0.1006)
2026-07-04 18:11:39	critical	0.1615	0.1006	Normal Sinus Rhythm detected (score=0.1615, confidence=0.1006)
2026-07-04 18:11:39	critical	0.1685	0.1006	Normal Sinus Rhythm detected (score=0.1685, confidence=0.1006)
2026-07-04 18:11:40	critical	0.1615	0.1006	Normal Sinus Rhythm detected (score=0.1615, confidence=0.1006)
2026-07-04 18:11:40	critical	0.1685	0.1006	Normal Sinus Rhythm detected (score=0.1685, confidence=0.1006)
2026-07-04 18:11:40	critical	0.1744	0.1006	Premature Atrial Contraction detected (score=0.1744, confidence=0.1006)
2026-07-04 18:11:40	warning	0.1506	0.1006	Premature Atrial Contraction detected (score=0.1506, confidence=0.1006)
2026-07-04 18:11:41	critical	0.1615	0.1006	Normal Sinus Rhythm detected (score=0.1615, confidence=0.1006)
2026-07-04 18:11:41	warning	0.1506	0.1006	Premature Atrial Contraction detected (score=0.1506, confidence=0.1006)
2026-07-04 18:11:41	critical	0.1615	0.1006	Premature Atrial Contraction detected (score=0.1615, confidence=0.1006)
2026-07-04 18:11:41	warning	0.1506	0.1006	Premature Atrial Contraction detected (score=0.1506, confidence=0.1006)
2026-07-04 18:11:41	critical	0.1615	0.1006	Premature Atrial Contraction detected (score=0.1615, confidence=0.1006)
2026-07-04 18:11:41	warning	0.1506	0.1006	Premature Atrial Contraction detected (score=0.1506, confidence=0.1006)

... and 312 more alerts not shown.

Detected Anomaly Waveforms

The following plots show the actual ECG segments (red) that triggered the most significant alerts, alongside the autoencoder's reconstruction of what it expected to see (blue dashed). The shaded region between the two curves represents the reconstruction error that exceeded the detection threshold. Arranged in descending order of anomaly score.



Clinical Summary

69.8% of analyzed samples were flagged as anomalous, suggesting sustained abnormal ECG morphology. Urgent cardiology review is recommended.

Physician Notes

Physician signature: _____

Date: _____

This report is generated automatically by an unsupervised deep-learning anomaly detection system and is intended for decision support only. It does not constitute a medical diagnosis and should not replace clinical judgment by a qualified physician.